

Mohammad Mobassir Ameen

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PROFILE

PhD researcher in Experimental High Energy Computational Physics at IIT Madras with strong foundations in Python, Machine Learning, and data-driven modeling. Experienced in applying ML techniques (BDTs, DNNs) to large-scale scientific datasets at CMS experiment, CERN.

SKILLS

Programming:	Python, C/C++, Bash/Shell Scripting, LaTeX/overleaf
Machine Learning & AI:	Supervised Learning, Binary & Multiclass Classification model like BDTs, DNNs, and Transformer based architecture model
Tools & Frameworks:	Git/GitHub, GitLab, Linux, ROOT, RooFit, CMSSW, RDataFrame
Data Handling:	Large-scale datasets, statistical analysis, visualization

WORK EXPERIENCE

PhD Researcher IIT Madras – CMS Experiment (CERN, Switzerland) Jul 2019 - present

- Developed ML-based classification models (BDT, DNN, and Transformer) for signal-background discrimination in multi-lepton final states.
- Designed Python-based data analysis pipelines for large-scale experimental datasets (Run-II & Run-III CMS data).
- Applied statistical learning methods to optimize feature selection and model performance.
- Worked collaboratively in an international research environment, contributing to analysis workflows and validations.

L3 Validation Manager – Physics Data & Monte Carlo Validation (PdmV) 2020 - 2022

- Validated large-scale production and pre-release datasets using automated Python workflows.
- Identified inconsistencies in data and MC using statistical checks and visualization.
- Collaborated with cross-functional teams to improve software reliability.

ttH + HH Monte Carlo Simulation Contact – HIG MC Team 2023 - 2025

- Coordinated Monte Carlo sample production and validation.
- Automated checks using Python scripts, improving turnaround time and consistency.

RESEARCH PROJECTS

Ph.D. Thesis Title: Studies of Higgs Boson Production in Association with Top Quarks in Multilepton Final States at CMS Experiment and Performance of Monolithic Pixel Sensor under X-ray Irradiation

Status: Thesis submitted; defense expected in August or September 2026.

EDUCATION

2019 - present	PhD (Physics) at Indian Institute of Technology, Madras	(CGPA: 8.0/10.0)
2016 - 2018	M.Sc. Physics at Aligarh Muslim University, Aligarh	(GPA: 7.29/10.0)
2012 - 2015	B.Sc. Physics at Aligarh Muslim University, Aligarh	(61.33%)

SCHOLASTIC ACHIEVEMENTS

2016	IIT JAM (Joint Admission Test for Masters)
2018	Graduate Aptitude Test in Engineering (GATE)
2018	CSIR-NET (December)

PUBLICATIONS

Journal Articles

A complete list of my journal publications with the CMS Collaboration can be found on [INSPIREHEP](#) or my [Google Scholar](#). Some selected journal publications are listed below:

- Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV, CMS Collaboration, arXiv:2602.18223 [hep-ex]
- M. M. Ameen et al., Performance of monolithic CMOS pixel sensors under X-rays, Submitted to Journal *Physics - Status Solidi*, **Status:** Accepted; it will publish as soon as the proofreading and editing have been completed
- Measurements of inclusive and differential Higgs boson production cross sections at $\sqrt{s} = 13.6$ TeV in the $H \rightarrow \gamma\gamma$ decay channel, CMS Collaboration, JHEP 09 (2025) 070 DOI: 10.1007/JHEP09(2025)070

In Conference Proceedings

Ameen, Mohammad Mobassir (2024). “Observation of the Decays $B^0 \rightarrow \psi(2S)K_S^0\pi^+\pi^-$ and $B_S^0 \rightarrow \psi(2S)K_S^0$ at $\sqrt{s} = 13$ TeV in the CMS Experiment”. In: *Proceedings of the XXV DAE-BRNS High Energy Physics (HEP) Symposium 2022*. Springer Nature Singapore, pp. 725–726.

Ameen, Mohammad Mobassir (2025). “Measurement of the ttH Production Cross-Section in Multi-leptonic Final States in pp Collisions at a Centre-of-Mass Energy of 13 TeV with the CMS Detector”. In: *Proceedings of the 17th International Conference on Interconnections Between Particle Physics and Cosmology*. Springer Nature Singapore, pp. 297–301.

WORKSHOPS & TRAINING

5-9 Feb 2024	CMS Week at CERN, Geneva, Switzerland
29 Jan-2 Feb 2024	6th Inter-experiment Machine Learning Workshop at CERN, Geneva, Switzerland
12-13 Feb 2024	CMS Physics Days: Ideas of Physics Analysis with ‘2022 and ‘2023 Data at CERN, Geneva, Switzerland
1-5 May 2023	HSF-India Training event at TIFR, Mumbai, India
12-25 Feb 2023	XV ICFA School on Instrumentation in Elementary Particle Physics at TIFR, Mumbai, India
4-14 Jan 2022	CMS Data Analysis School by the LHC Physics Center at Fermilab, USA (Virtually)

HONORS & AWARDS

- Breakthrough Prize in Fundamental Physics (shared with colleagues in the CMS Collaboration)
- Best Poster Award, DESY (Germany) – PO & Data Analysis School 2023
- Advance Research Opportunities Program (AROP) Research Fellowship to visit RWTH Aachen University, Germany (for 6 months)
- International Immersion Experience (IIE) Travel Award, IIT Madras
- Half-time Research Assistantship (HTRA), July 2019 - July 2024

MENTORING

- Jan-May 2026: Mentored a M.Sc. Physics (IITM) on the project - “Identification of Jets Originating from the Higgs Boson Using Binary Classification Models”
- May-July 2026 : Mentored three M.Sc. Physics intern students from University of Madras, Guided them in high-energy physics data analysis software tools, ROOT/Python, and machine learning applications.

TEACHING ASSISTANTSHIP

- EP3291: Engineering Physics Lab IV, Department of Physics, IIT Madras
- PH1010: PHYSICS I : Classical Mechanics, Department of Physics, IIT Madras
- PH1020: PHYSICS II : Electricity & Magnetism, Department of Physics, IIT Madras

HOBBIES

- Traveling (exploring new cities/country), interacting with the researcher to learn more